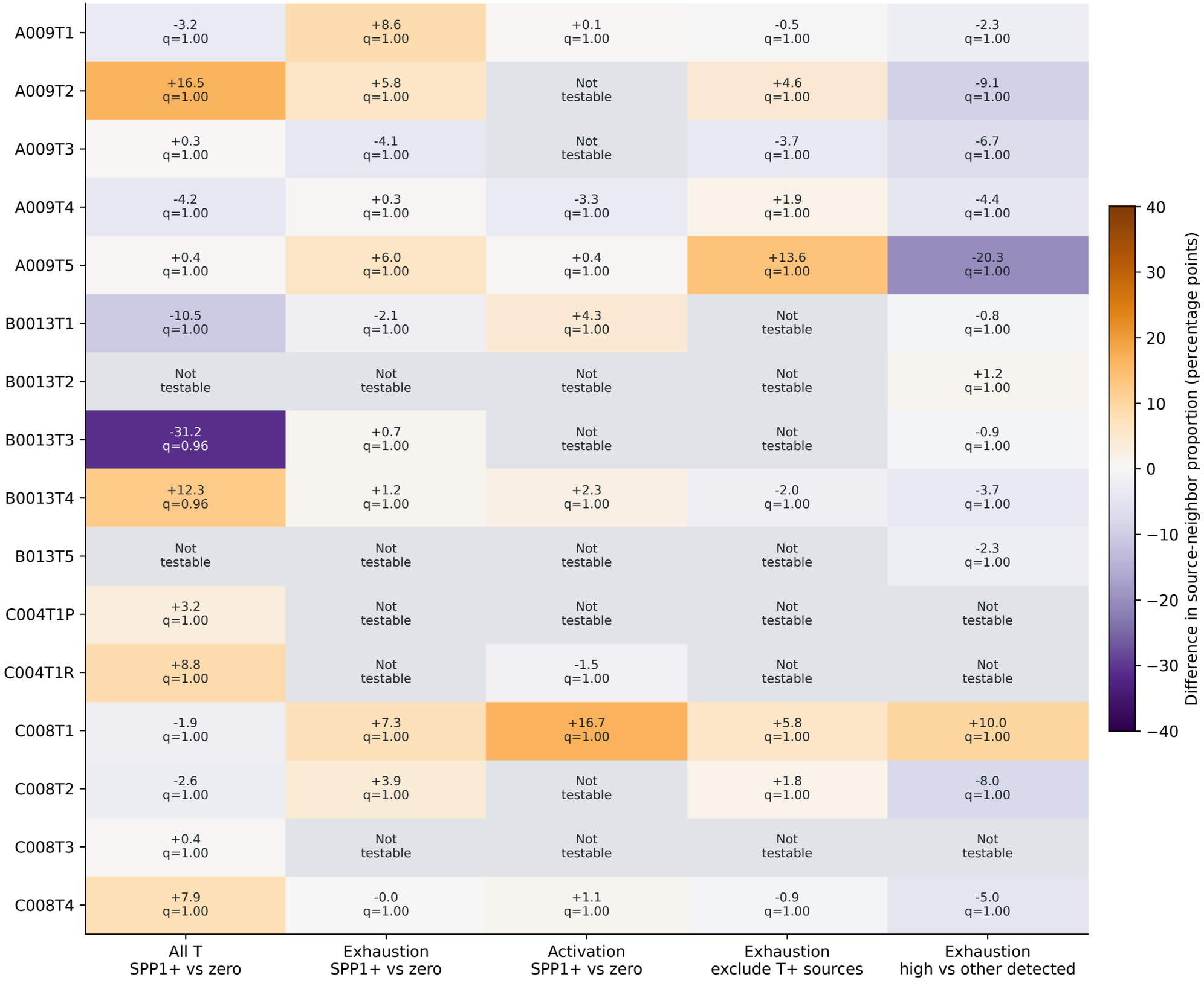

q: BH across 176 planned tests; 4,999 within-tile/UMI/Mac-core permutations. Grey: missing comparison or insufficient/constant data.
Spot-level conditional association; sections are not independent patients. Positive: greater T adjacency / stronger neighbor program.